Sikai Cheng

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Education

Georgia Institute of Technology — **Aug. 2024 – Present**
Ph.D. in Machine Learning, Advisor: Pascal Van Hentenryck — *Atlanta, GA*

Georgia Institute of Technology — **Aug. 2022 – Dec. 2023**
M.S. in Operations Research (GPA: 3.91/4.00) — *Atlanta, GA*

The Chinese University of Hong Kong — **Aug. 2018 – May 2022**
B.S. in Statistics, Data Science Stream (Major GPA: 3.92/4.00; Rank: 1/199) — *Shenzhen, China*

Research Interests

My research focuses on **agentic AI**, particularly using LLM agents for **algorithm evolution** and **recursive self-improvement**, and on **benchmarking** what agents can do. I also study the efficiency and reliability of **language models**. In addition, I have worked on **ML-integrated optimization** for **large-scale logistics**, **deep time-series forecasting** for **5G wireless communication**, and **multi-agent reinforcement learning**.

Skills

- **Programming and ML:** Python (PyTorch/Lightning, scikit-learn, XGBoost, LightGBM, pandas), Gurobi, SQL, C++, CUDA(C)
- **Systems and Tools:** AWS (S3, SageMaker, EC2), Slurm HPC clusters, Git, LaTeX, Bash, Linux, Hugging Face

Selected Publications

Agentic AI

1. Jiang, H., Guan, C., Li, M., Li, D., **Cheng, S.**, Van Hentenryck, P., & Bent, R. (2026). **OPTAgent: From Standalone Solvers to Deployment-Aware Solver Ecosystems.** *MLxOR Workshop at NeurIPS 2026*. [PDF]
2. Li, X., Liu, Y., Chen, W., You, B., Di, Z., …, **Cheng, S.**, …, Song, D. (2026). **SkillsBench: Benchmarking How Well Agent Skills Work Across Diverse Tasks.** *NeurIPS 2026 (Evaluations & Datasets Track)*. [PDF] [Code]

Language Models

3. Li, M., Jiang, H., **Cheng, S.**, Fu, H., Cai, Y., Huang, B., Chen, Z., Ye, T., Chen, X., & Van Hentenryck, P. (2026). **Plan, Verify and Fill: A Structured Parallel Decoding Approach for Diffusion Language Models.** *Findings of the Association for Computational Linguistics: EMNLP 2026*. *Spotlight, NonAR-LM Workshop at COLM 2026*. [PDF]
4. Bruys, T., **Cheng, S.**, Moradi, A., Van Hentenryck, P., & Zandehshahvar, R. (2026). **Toward Reliable Generative AI with Conformal Prediction: A Survey.** *Manuscript submitted for publication*. [PDF] [Code]
5. Klamkin, M., Deza, A., **Cheng, S.**, Zhao, H., & Van Hentenryck, P. (2025). **DualSchool: How reliable are LLMs for optimization education?** *arXiv preprint arXiv:2505.21775*. [PDF]

ML-Integrated Optimization

6. **Cheng, S.**, Hijazi, A., Konak, J., Erera, A., & Van Hentenryck, P. (2025). **SPOT: Spatio-Temporal Pattern Mining and Optimization for Load Consolidation in Freight Transportation Networks.** *Proceedings of the IEEE International Conference on Data Mining (ICDM), 2025*. (**Acceptance rate: 13%**). [PDF]
7. Ye, T., **Cheng, S.**, Hijazi, A., & Van Hentenryck, P. (2025). **Contextual stochastic optimization for omnichannel multi-courier order fulfillment under delivery time uncertainty.** *Manufacturing & Service Operations Management (M&SOM)*. [PDF]

Deep Learning for Time Series and 5G Wireless Communication

8. **Cheng, S.**, Zandehshahvar, R., Zhao, H., Garcia-Ulloa, D. A., Villena-Rodriguez, A., Navarro Manchón, C., & Van Hentenryck, P. (2026). **CSI-4CAST: A Hybrid Deep Learning Model for CSI Prediction with Comprehensive Robustness and Generalization Testing.** *IEEE Transactions on Machine Learning in Communications and Networking*, vol. 4, pp. 882–912. doi: 10.1109/TMLCN.2026.3694830. [PDF] [Code] [Dataset]

Works in Progress

- **Cheng, S.**, Ye, T., Li, M., et al. **EvoDecomp: Agent-Evolved Decomposition for Large-Scale Freight Routing.**
- **Cheng, S.**, Ye, T., Li, M., et al. **DecompBench: Can Coding Agents Write Expert-Grade Decomposition Solvers?**

Experience

Amazon **May 2026 – Aug. 2026**

Applied Scientist Intern *Seattle, WA*

- Built a causal evaluation framework for product-substitution recommendations using difference-in-differences on a randomized field experiment across ∼1,000 treatment and control locations, with FDR correction over a large target–candidate search space.
- Developed product representations from a generative customer-behavior foundation model; across six aggregation strategies a sampled mean matched costlier alternatives, improving recovered demand by 3–5% over the production baseline.
- Designed a counterfactual sequence-likelihood signal from the same model, scaling evaluation over 2.5 TB of event data through 200+ AWS SageMaker batch-inference jobs (∼700 GPU-hours).
- Redesigned candidate generation using functional similarity, improving recovered demand by 10–15% at a fixed selection budget.
- Prototyped an AI-agent system identifying substitutes directly from product detail pages, doubling recovered demand over the existing method on a small difficult-case pilot.

AI Institute for Advances in Optimization, Georgia Institute of Technology **Aug. 2024 – Present**

Graduate Research Assistant *Atlanta, GA*

- Deep Learning for Time Series Forecasting — Project with a Leading Global Wireless Communications Company
  - Proposed lightweight yet expressive deep learning models for channel state information (CSI) prediction in 5G systems, demonstrating superior robustness and generalization across thousands of realistic scenarios while reducing computational cost by up to 5× compared to the SOTA LLM-based predictor.
- AI-Integrated Optimization — Projects with a Fortune 500 Logistics Company and a Top U.S. Retailer
  - Developed an integrated spatio-temporal pattern-mining and optimization framework for freight-load consolidation, reducing transportation costs and travel distance by ∼50% on large-scale industrial datasets; framework deployed in real operations.
  - Developed a contextual stochastic optimization framework that integrates distributional delivery-time forecasts with robust order fulfillment models, providing actionable insights that improved service quality and customer satisfaction.
- Large Language Models and Generative AI
  - Investigated limitations in the reasoning and mathematical understanding of large language models by co-developing a novel benchmark for linear-program dual generation, thereby revealing fundamental gaps in model reliability.
  - Surveyed conformal prediction methods for generative AI and developed a unified taxonomy spanning open-ended generation, factuality, RAG, multimodal systems, and structured outputs.
  - Co-constructed a training-free decoding method for diffusion language models that verifies informative low-confidence tokens, reducing forward passes by 43–65% and increasing throughput by up to 2.3× while maintaining comparable accuracy.

S.F. Express Co., Ltd. **May 2023 – Aug. 2023**

Research Intern *Shenzhen, China*

- Developed a novel approach for transportation tasks by formulating them as a variant of the Capacitated Vehicle Routing Problem with time limitations (CVRP); implemented Tabu Search Heuristic and Gurobi-based MIP solver, improving profits by 10%.
- Integrated inventory and routing decisions into an Inventory Routing Problem (IRP); applied Lagrangian Relaxation Methods and Fix-Partition-Policy to optimize performance.
- Modeled transportation tasks as a graph-to-sequence problem; designed GCN encoder and pointer network decoder, achieving near-optimal solutions on real-world datasets.

Parametrix Technology Co., Ltd. **Dec. 2021 – Apr. 2022**

Research Intern *Shenzhen, China*

- Developed multi-agent reinforcement learning models for the Lux.AI Kaggle competition, enabling cooperation among agents through neighborhood-based and grid-based modeling.
- Implemented multi-stage training with knowledge distillation to leverage both dense and sparse reward signals.
- Built and deployed evaluation environments; the final model achieved a win rate >90% against top-ranked Kaggle agents.
- Co-authored the paper *Emergent Collective Intelligence from Massive-Agent Cooperation and Competition*, later presented at NeurIPS 2022 Deep RL Workshop.

Awards

- Honorary Mention, M&SOM Practice-Based Research Competition, 2025
- ICDM Student Travel Award, 2025
- Undergraduate Research Award (2021, 2022) and Academic Achievement Scholarship (2020, 2021), The Chinese University of Hong Kong